BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 9

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Subject Name : Deep Learning

1. Aim

To perform image classification using a pre-trained Vision Transformer (ViT-B/16) and compare its performance with a pre-trained CNN (ResNet18) using a small image dataset.

2. Objectives

Understand transfer learning; implement ViT and CNN classifiers; compare test accuracy, loss, and runtime; use a small dataset to reduce data-download requirements.

3. Software and Hardware Requirements

Python 3.10/3.11, VS Code, PyTorch, Torchvision, scikit-learn, NumPy, Pillow, and Matplotlib. A CUDA-enabled GPU is helpful but not required. Internet is required on the first run to download pretrained model weights.

4. Dataset

This practical uses scikit-learn's built-in Digits dataset, containing 1,797 small 8×8 grayscale images representing digits 0–9. The dataset is bundled with scikit-learn and does not require downloading a separate image archive. Images are converted to RGB and resized to 224×224 for the pretrained models. Default experiment uses 800 training samples and 300 test samples.

5. Theory

A Vision Transformer splits an image into patches and uses self-attention to model relationships among them. A CNN uses convolutional filters to learn spatial patterns. Transfer learning reuses ImageNet-pretrained features. This implementation replaces each model's output head with a 10-class head and freezes the backbone to reduce training cost. The pretrained model files themselves are still downloaded on first use and may be large.

6. Algorithm / Procedure

1. Create and activate a Python virtual environment.\n2. Install requirements.txt.\n3. Load the built-in Digits dataset and choose reproducible train/test subsets.\n4. Convert each grayscale image to RGB, resize to 224×224, and normalize.\n5. Load pretrained ViT-B/16 and ResNet18; replace each final classifier with a 10-class

head.\n6. Train the new heads with identical data splits and settings.\n7. Evaluate test accuracy and loss and record runtime.\n8. Save CSV metrics and a comparison plot.

7. Execution Commands

Windows PowerShell:\npy -3.11 -m venv .venv\n.\\.venv\\Scripts\\Activate.ps1\npython -m pip install --upgrade pip\npip install -r requirements.txt\npython main.py\n\nLightweight run (CNN only): python main.py --models cnn --train-samples 200 --test-samples 100 --batch-size 4. To compare both: python main.py --train-samples 400 --test-samples 150 --batch-size 4.

8. Results

Fill the table using outputs/comparison_results.csv after executing the program. Do not invent metrics.

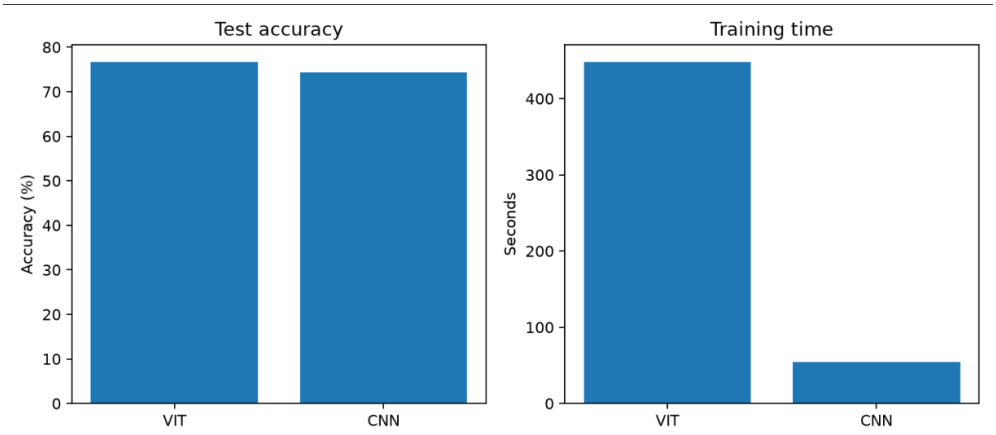
Model	Test Accuracy (%)	Test Loss	Train loss
ViT-B/16	_76.67%	0.9171	1.4202
ResNet18 (CNN)	_74.33%_	1.2651	1.8947

9. Screenshot of output

Screenshot 1: ViT and CNN model execution output.

```
(.venv) PS C:\Sem5-(TY)\Deep Learning\Deep Learning Labs\DL_Practical_01_ViT_vs_CNN> python main.py
Device: cpu
Dataset: built-in sklearn Digits (no separate dataset download).
First run downloads pretrained model weights only.
\n================ VIT ================
Downloading: "https://download.pytorch.org/models/vit_b_16-c867db91.pth" to C:\Users\ThinkPad/.cache\torch\hub\checkpoints
\vit_b_16-c867db91.pth
100.0%
Epoch 1/1 | train loss 1.4202 | test loss 0.9171 | test accuracy 76.67%
\n================ CNN ================
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to C:\Users\ThinkPad/.cache\torch\hub\checkpoints
\resnet18-f37072fd.pth
100.0%
Epoch 1/1 | train loss 1.8947 | test loss 1.2651 | test accuracy 74.33%
\nFINAL RESULTS
{'model': 'VIT', 'test_accuracy_percent': 76.67, 'test_loss': 0.9171, 'training_seconds': 448.08}
{'model': 'CNN', 'test_accuracy_percent': 74.33, 'test_loss': 1.2651, 'training_seconds': 54.7}
Saved results in outputs/.
(.venv) PS C:\Sem5-(TY)\Deep Learning\Deep Learning Labs\DL_Practical_01_ViT_vs_CNN>
```

Screenshot 2: Generated comparison plot and CSV results.



10. Discussion

Compare observed accuracy, test loss, and runtime. Results depend on the selected subset, pretrained features, hardware, and number of epochs. The Digits dataset is intentionally small and simple; this experiment demonstrates implementation and comparison rather than establishing general superiority of either architecture.

11. Conclusion

The practical demonstrates image classification with a pre-trained ViT and CNN using a small dataset that does not require downloading a large image archive. Complete this conclusion with the measured results from your own execution.

12. References

PyTorch: https://pytorch.org/

Torchvision models: https://pytorch.org/vision/stable/models.html

scikit-learn Digits dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html

Dosovitskiy et al., An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, ICLR 2021.